

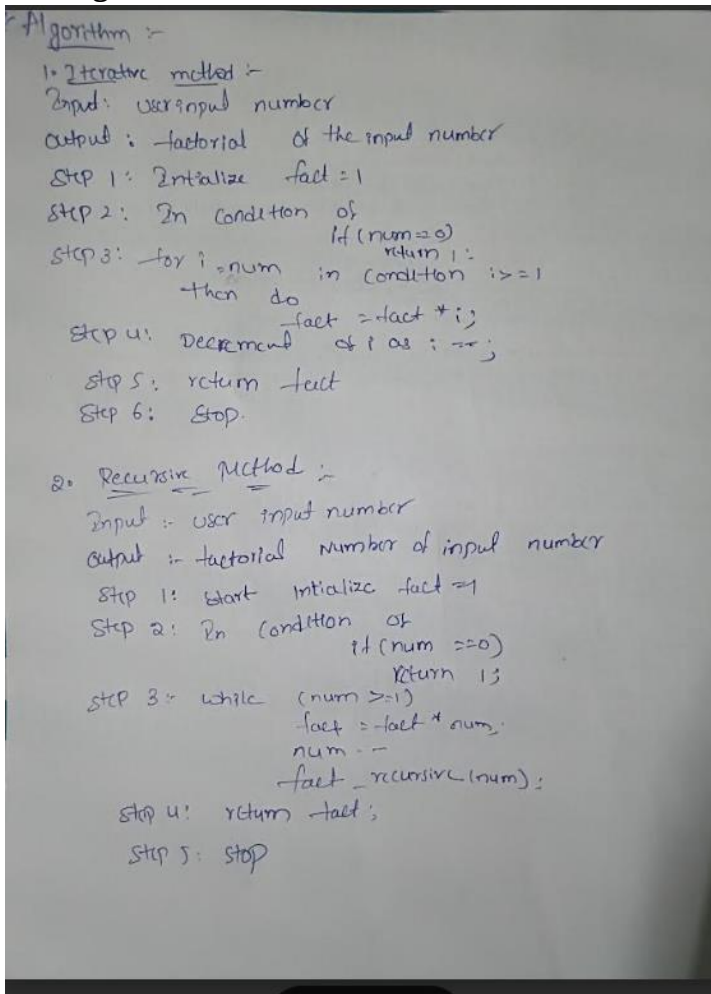
Practical: 4

Implementation and Time analysis of factorial program using iterative and recursive method

18/08/2026

Aim: To Implement and Time analysis of factorial program using iterative and recursive method.

Algorithm:



Program:

factorial program using iterative and recursive method

```
#include<iostream>
using namespace std;
```

```
int fact_iterative(int num){
    int fact = 1;
    if(num == 0)
        return 1;
    for(int i=num; i>=1; i--){
        fact = fact*i;
    }
    return fact;
}

int fact_recursive(int num){
    int fact = 1;
    if(num == 0)
        return 1;
    while(num >= 1){
        fact = fact*num;
        num--;
        fact_recursive(num);
    }
    return fact;
}

int main(){
    cout<<"-----Factorial-----"<<endl;
    int n;
    cout<<"Enter your number: ";
    cin>>n;
    int i = fact_iterative(n);

    cout<<"Iterative method:"<<endl;
    cout<<n<<" factorial is "<<i;
    cout<<endl<<endl;
    int r = fact_recursive(n);
    cout<<"Recursive method:"<<endl;
    cout<<n<<" factorial is "<<r;

    return 0;
}
```

Output:

Factorial

Enter your number: 6

Iterative method:

6 factorial is 720

Recursive method:

6 factorial is 720

=== Code Execution Successful ===

Time Complexity:

Time Complexity:-

① Iterative method:-

```

fact = 1; → ①
if (num == 0) → 1
    return 1 → 1
for (i = num; i >= 1; i--) { → 1+num
    fact = fact * i; → n
}
return fact; → 1
    
```

Input: 6
 Output: 720
 The Time complexity is $O(n)$

Best case $O(1)$
 Average $O(n)$
 Worst $O(n)$
 Space complexity $O(1)$

② Recursive Method:-

```

int fact(int num) { → ①
    int fact = 1; → 1
    if (num == 0) → 1
        return 1 → 1
    while (num >= 1) { → n
        fact = fact * num; → n
        num--; → n
    }
    fact(num); → 1
    return fact; → 1
}
    
```

Input: 6
 Output: 720
 Time complexity is $O(n)$
 Space complexity is $O(1)$

Best case $O(1)$
 Average $O(n)$
 Worst $O(n)$



Conclusion:

The factorial program was successfully implemented using iterative and recursive methods.

Both methods produce the correct factorial result.

The program demonstrates the difference between iteration and recursion.